

Precision Strain Gauge Weight Scale

Op-Amp Circuit Design, Simulink Verification & MATLAB DAQ

OBJECTIVE

Design, build, and program a complete strain-gauge weight scale from raw millivolt sensor signal through analog conditioning to a calibrated real-time mass readout in MATLAB.

MY CONTRIBUTION

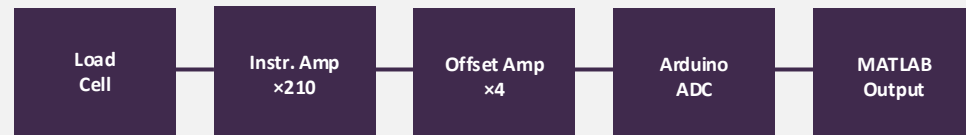
Instrumentation amplifier: full V_{out} derived via nodal analysis

Simulink verification: circuit modelled to confirm gain before physical build; $V_{out} = 1.45\text{ V}$ and gain = 210 confirmed

Offset amplifier: solved 2×2 resistor system to scale output to Arduino ADC range (0.5–2.5 V)

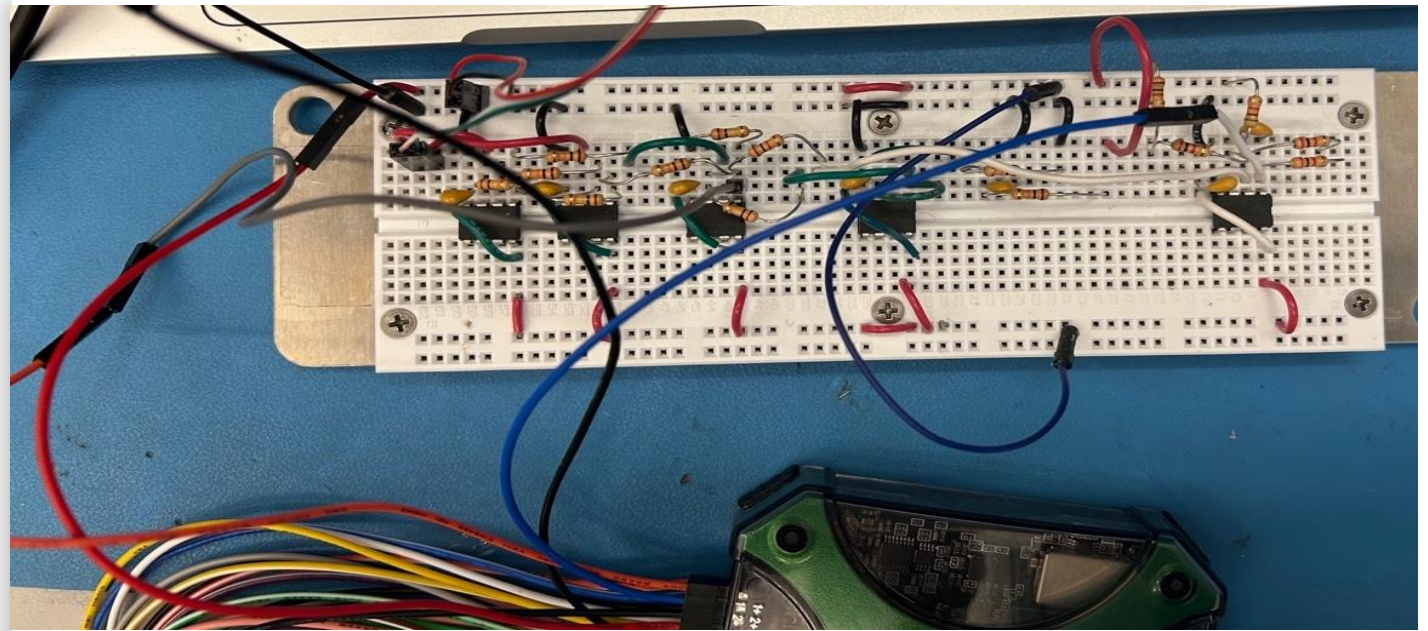
MATLAB DAQ program: 100 Hz acquisition loop; calibrated equation mass = $1.11\text{ V} - 0.591$ validated at 0 & 2 kg

SIGNAL CHAIN

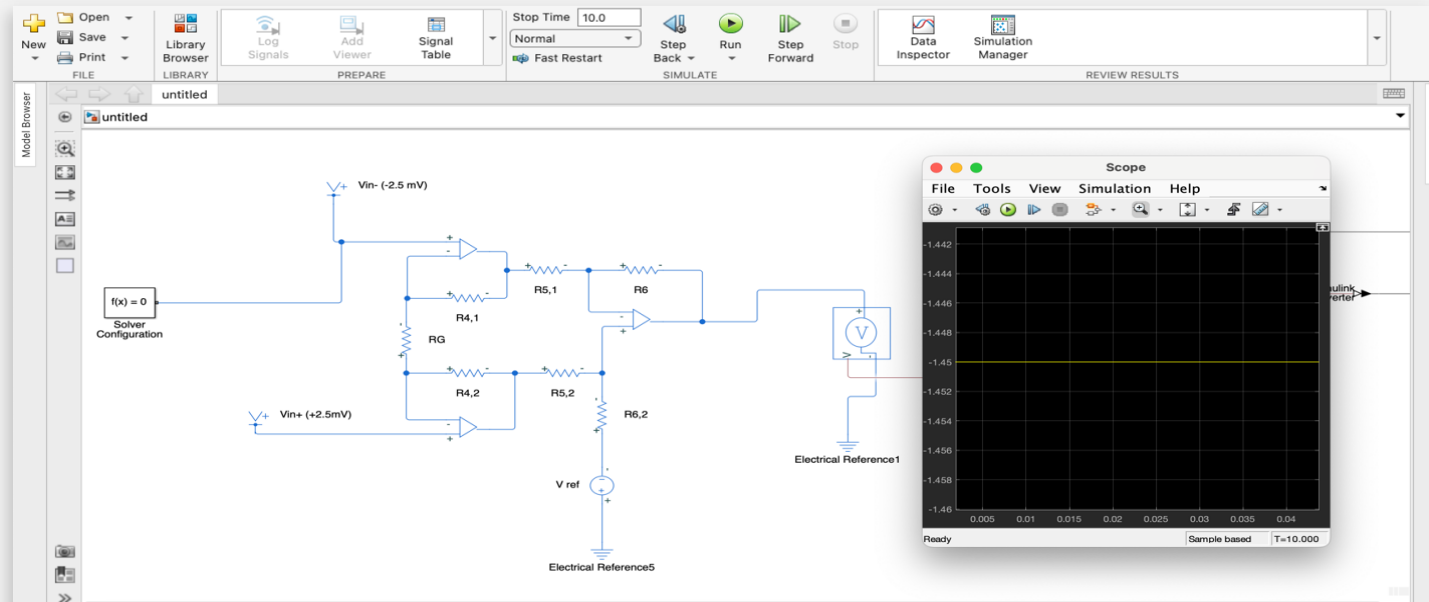


OUTCOME

Measured gain = 210. Output 0.62–2.41 V (target 0.5–2.5 V). Mass readout validated at 0 kg & 2 kg. Minor deviation from standard resistor value substitutions.



Assembled instrumentation + offset amplifier circuit on breadboard with function generator



Simulink model verifying gain = 210 before physical breadboard build